

Match the *most appropriate* term to each concept. Not all terms are used and no term is used twice. (2 points each)

|                           |                            |                 |
|---------------------------|----------------------------|-----------------|
| Annual Harvestable Growth | Finfish                    | Microbial Loop  |
| Bioconcentration          | Fisheries Equilibrium      | Net Plankton    |
| Bioluminescence           | Flippers                   | Phosphorescence |
| Biomagnification          | Interference Competition   | Photophores     |
| Bloom                     | Light Competition          | Phytoplankton   |
| Carrying Capacity         | Logistic Population Growth | Scales          |
| Commercial Equilibrium    | Luminators                 | Scutes          |
| Epiplankton               | Marmalade                  | Shellfish       |
| Exploitative Competition  | Maximum Sustainable Yield  | Wave Action     |
| Explosive Growth          | Meroplankton               | Zooplankton     |

- \_\_\_\_\_ Light-emitting organs on deep-sea organisms.
- \_\_\_\_\_ Balance between harvest and population growth rate.
- \_\_\_\_\_ Limits amount of DOM that is exported from the epipelagic to the deep sea.
- \_\_\_\_\_ A commercial fishery consisting of arthropods and molluscs.
- \_\_\_\_\_ For a long time, the only type of plankton known from the epipelagic.
- \_\_\_\_\_ Increasing accumulation of toxins by organisms in higher trophic levels.
- \_\_\_\_\_ The lack of these separates leatherback turtles from other sea turtles.
- \_\_\_\_\_ Rapid increase in abundance of jellyfish in a small area.
- \_\_\_\_\_ Rapid coral growth blocks light from slower growing species.
- \_\_\_\_\_ What is your favorite movie? (not in list above so get it right!)

(5 pts) 11. Briefly explain two reasons why coral reefs are not found near large estuaries in the tropics.

(5 pts) 12. Describe one specific example of how a deep-sea organism uses bioluminescence.

(5 pts) 13. The deepest regions of the bathypelagic zone have less than 1% of the biomass that is found in zones directly above it. Explain one reason why.

- (5 pts) 14. (a) On screen is a whipnose seadevil (genus *Gigantactis*). State whether the whipnose seadevil more likely lives in the mesopelagic or bathypelagic zone, and explain one adaptation, *visible in the image*, the fish exhibits to justify your answer. Be sure to explain the specific relationship between the adaptation and the zone you choose.
- (5 pts) (b) Identify a different adaptation, *visible in the image*, that is specific to life in the deep-sea. Explain how that adaptation is related to the environment or survival in the deep-sea.

(5 pts) 15. (a) Illustrate a profile of a barrier reef from shoreline to open ocean. Label each zone.

(10 pts) (b) For each zone you just identified, describe a representative coral shape or structure you will find in that zone. Relate the shape or structure of the coral to one primary physical process (e.g., light, temperature, tidal movement, turbidity, wave action, or others, as appropriate) present in each zone.

- (15 pts) 16. Southern Florida and coastal Louisiana once had extensive salt marshes (and mangrove forests in the case of Florida). Explain two different ways that humans have caused the loss of these extensive marshes. Then, explain the potential impact that the loss of these valuable estuaries could have on the associated marine communities and on humans.

2 pts extra credit: List, in order from smallest to largest, the seven size classes of plankton. Do not list the sizes of each class. Must be exact order to receive credit.